

Team "A for Apple" — ChemE 486 Process

Team Members: Alpha Arestun · Mark Heien · Angelina Pa

Task

Team setup & orientation

Literature review

Report #1 – Background & motivation
(Apple pomace / Ethanol / Fermentation)

Coal gasification Aspen model (Individual – Report #2)

Fermentation process design
(apple pomace → ethanol flowsheet)

Draft figure preparation

Retrofitted plant Aspen model – Report #3 (Team)

Economic analysis (CAPEX / OPEX / NPV)

Life cycle assessment & P&ID development

Oral Report #1 – Technical presentation

Oral Report #2 – Non-technical short-form video

Final report writing

Key Milestones



Process Design II, Spring 2026 | Feedstock: Apple Pomace · Product: Ethanol

Information · Lorrie Welch

Phase	
Background & Writing	
Background & Writing	
Background & Writing	
Coal Gasification Model	
Retrofitted Plant Design	
Retrofitted Plant Design	
Retrofitted Plant Design	
Analysis	
Analysis	
Communications & Final Rpt	
Communications & Final Rpt	
Communications & Final Rpt	

Deliverable	Due Date
Report #1 – Background & motivation (Apple pomace / Ethanol / Fermentation)	Friday, April 17, 2026
Coal gasification Aspen model (Individual – Report #2)	Friday, May 1, 2026
Draft figure preparation	Tuesday, May 12, 2026
Retrofitted plant Aspen model – Report #3 (Team)	Friday, May 15, 2026
Oral Report #1 – Technical presentation	Tuesday, May 26, 2026
Oral Report #2 – Non-technical short-form video	Tuesday, June 2, 2026
Final report writing	Friday, June 5, 2026

[illegible]

Type		
Team Written		
Individual Written		
Lab Deliverable		
Team Written		
Team Oral		
Team Oral (Video)		
Team Written		

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